

Formula Sheet for Vector Analysis and Complex Variables MAT2238

1 Grad, Div, Curl

All vectors of the form $\mathbf{a} = [a_1, a_2, a_3] = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ constitute the **real vector space** $\mathbb{R}^3$ with componentwise vector addition

$$[a_1, a_2, a_3] + [b_1, b_2, b_3] = [a_1 + b_1, a_2 + b_2, a_3 + b_3] \quad (1)$$

and componentwise scalar multiplication (c a scalar, a real number)

$$c[a_1, a_2, a_3] = [ca_1, ca_2, ca_3] \quad (2)$$

The **inner product** or **dot product** of two vectors is defined by

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \gamma = a_1b_1 + a_2b_2 + a_3b_3 \quad (3)$$

where γ is the angle between $\mathbf{a}$ and $\mathbf{b}$. This gives for the **norm** or **length** $|\mathbf{a}|$ of $\mathbf{a}$

$$|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2} \quad (4)$$

as well as a formula for γ . If $\mathbf{a} \cdot \mathbf{b} = 0$, we call $\mathbf{a}$ and $\mathbf{b}$ **orthogonal**.

The **vector product** or **cross product** $\mathbf{v} = \mathbf{a} \times \mathbf{b}$ is a vector of length

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \gamma \quad (5)$$

and perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ such that $\mathbf{a}, \mathbf{b}, \mathbf{v}$ form a *right-handed* triple. In terms of components with respect to right-handed coordinates,

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (6)$$

We extend differential calculus to vector valued functions

$$\mathbf{v}(t) = [v_1(t), v_2(t), v_3(t)] = v_1(t)\mathbf{i} + v_2(t)\mathbf{j} + v_3(t)\mathbf{k}$$

The derivative of $\mathbf{v}(t)$ is

$$\mathbf{v}' = \frac{d\mathbf{v}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{v}(t + \Delta t) - \mathbf{v}(t)}{\Delta t} = [v'_1, v'_2, v'_3] = v'_1\mathbf{i} + v'_2\mathbf{j} + v'_3\mathbf{k}. \quad (8)$$

Curves C in space represented by the position vector $\mathbf{r}(t)$ have $\mathbf{r}'(t)$ as a **tangent vector** (the *velocity* in mechanics when t is time), $\mathbf{r}'(s)$ (s arc length) as the *unit tangent vector*.

Tangent Line to a Curve: For a curve $\mathbf{r}(t)$, the equation of the tangent line at $t = t_0$ is:

$$\mathbf{L}(\omega) = \mathbf{r}(t_0) + \omega\mathbf{r}'(t_0)$$

We also extend differential calculus to vector valued functions of more than one variable. **Vector valued functions** $\mathbf{v}(x, y, z) = [v_1(x, y, z), v_2(x, y, z), v_3(x, y, z)]$ represent vector

fields in space. Partial derivatives with respect to the Cartesian coordinates x, y, z are obtained componentwise, for instance,

$$\frac{\partial \mathbf{v}}{\partial x} = \left[\frac{\partial v_1}{\partial x}, \frac{\partial v_2}{\partial x}, \frac{\partial v_3}{\partial x} \right] = \frac{\partial v_1}{\partial x} \mathbf{i} + \frac{\partial v_2}{\partial x} \mathbf{j} + \frac{\partial v_3}{\partial x} \mathbf{k}$$

The **gradient** of a scalar function f is

$$\text{grad } f = \nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right] \quad (9)$$

The **directional derivative** of f in the direction of a vector $\mathbf{a}$ is

$$D_{\mathbf{a}} f = \frac{df}{ds} = \frac{1}{|\mathbf{a}|} \mathbf{a} \cdot \nabla f \quad (10)$$

The **divergence** of a vector function $\mathbf{v}$ is

$$\text{div } \mathbf{v} = \nabla \cdot \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \quad (11)$$

The **curl** of $\mathbf{v}$ is

$$\text{curl } \mathbf{v} = \nabla \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} \quad (12)$$

or minus the determinant if the coordinates are left-handed.

Conservative Vector Fields: A vector field $\mathbf{F}$ is conservative if there exists a scalar potential ϕ such that $\mathbf{F} = \nabla \phi$. A necessary and sufficient condition (in a simply connected domain) is that $\text{curl } \mathbf{F} = \mathbf{0}$.

Some basic formulas for grad, div, curl are

$$\nabla(fg) = f\nabla g + g\nabla f \quad (13)$$

$$\nabla(f/g) = (1/g^2)(g\nabla f - f\nabla g)$$

$$\text{div}(f\mathbf{v}) = f\text{div } \mathbf{v} + \mathbf{v} \cdot \nabla f \quad (14)$$

$$\text{div}(f\nabla g) = f\nabla^2 g + \nabla f \cdot \nabla g$$

$$\nabla^2 f = \text{div}(\nabla f) \quad (15)$$

$$\nabla^2(fg) = g\nabla^2 f + 2\nabla f \cdot \nabla g + f\nabla^2 g$$

$$\text{curl}(f\mathbf{v}) = \nabla f \times \mathbf{v} + f\text{curl } \mathbf{v} \quad (16)$$

$$\text{div}(\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot \text{curl } \mathbf{u} - \mathbf{u} \cdot \text{curl } \mathbf{v}$$

$$\text{curl}(\nabla f) = \mathbf{0} \quad (17)$$

$$\text{div}(\text{curl } \mathbf{v}) = 0.$$

2 Complex Numbers and Functions. Complex Differentiation

For arithmetic operations with **complex numbers**

$$z = x + iy = re^{i\theta} = r(\cos \theta + i \sin \theta), \quad (1)$$

$$r = |z| = \sqrt{x^2 + y^2}, \quad \theta = \arctan(y/x).$$

A complex function $f(z) = u(x, y) + iv(x, y)$ is **analytic** in a domain D if it has a **derivative**

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad (2)$$

everywhere in D . Also, $f(z)$ is *analytic at a point* $z = z_0$ if it has a derivative in a neighborhood of z_0 (not merely at z_0 itself).

If $f(z)$ is analytic in D , then $u(x, y)$ and $v(x, y)$ satisfy the (very important!) **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (3)$$

everywhere in D . Then u and v also satisfy **Laplace’s equation**

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0 \quad (4)$$

everywhere in D . If $u(x, y)$ and $v(x, y)$ are continuous and have *continuous* partial derivatives in D that satisfy (3) in D , then $f(z) = u(x, y) + iv(x, y)$ is analytic in D .

The **complex exponential function**

$$e^z = \exp z = e^x(\cos y + i \sin y) \quad (5)$$

reduces to e^x if $z = x$ ($y = 0$). It is periodic with $2\pi i$ and has the derivative e^z .

The **trigonometric functions** are

$$\begin{aligned} \cos z &= \frac{1}{2}(e^{iz} + e^{-iz}) = \cos x \cosh y - i \sin x \sinh y \\ \sin z &= \frac{1}{2i}(e^{iz} - e^{-iz}) = \sin x \cosh y + i \cos x \sinh y \end{aligned} \quad (6)$$

and, furthermore,

$$\tan z = (\sin z)/\cos z, \quad \cot z = 1/\tan z, \quad \text{etc.}$$

The **hyperbolic functions** are

$$\cosh z = \frac{1}{2}(e^z + e^{-z}) = \cos iz, \quad \sinh z = \frac{1}{2}(e^z - e^{-z}) = -i \sin iz \quad (7)$$

etc. The functions (5)–(7) are **entire**, that is, analytic everywhere in the complex plane.

The **logarithm** function is defined as

$$\log z = \ln |z| + i \arg z = \ln |z| + i \operatorname{Arg} z \pm 2n\pi i \quad (8)$$

where $z \neq 0$ and $n = 0, 1, \dots$. $\operatorname{Arg} z$ is the **principal value** of $\arg z$, that is, $-\pi < \operatorname{Arg} z \leq \pi$. We see that $\log z$ is infinitely many-valued. Taking $n = 0$ gives the **principal value** $\operatorname{Log} z$ of $\ln z$; thus $\operatorname{Log} z = \ln |z| + i \operatorname{Arg} z$.

The **complex line integral** of a function $f(z)$ taken over a path C is denoted by

$$\int_C f(z) dz \quad \text{or, if } C \text{ is closed, also by} \quad \oint_C f(z) \quad (1)$$

If $f(z)$ is analytic in a simply connected domain D , then we can evaluate (1) as in calculus by indefinite integration and substitution of limits, that is,

$$\int_C f(z) dz = F(z_1) - F(z_0) \quad [F'(z) = f(z)] \quad (2)$$

for every path C in D from a point z_0 to a point z_1 . These assumptions imply **independence of path**, that is, (2) depends only on z_0 and z_1 (and on $f(z)$, of course) but not on the choice of C . The existence of an $F(z)$ such that $F'(z) = f(z)$ is proved by Cauchy's integral theorem. A general method of integration, not restricted to analytic functions, uses the equation $z = z(t)$ of C , where $a \leq t \leq b$,

$$\int_C f(z) dz = \int_a^b f(z(t)) \dot{z}(t) dt \quad \left(\dot{z} = \frac{dz}{dt} \right). \quad (3)$$

Cauchy's integral theorem is the most important theorem in this chapter. It states that if $f(z)$ is analytic in a simply connected domain D , then for every closed path C in D ,

$$\oint_C f(z) dz = 0. \quad (4)$$

Under the same assumptions and for any z_0 in D and closed path C in D containing z_0 in its interior we also have **Cauchy's integral formula**

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz. \quad (5)$$

Furthermore, under these assumptions $f(z)$ has derivatives of all orders in D that are themselves analytic functions in D and

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n = 1, 2, \dots). \quad (6)$$

Liouville's theorem states that an entire function that is bounded in the whole complex plane must be constant.

Important Theorems

I. VECTOR CALCULUS:-

1. Equality of partial derivatives :-

Two mixed partial derivatives $\frac{\partial^2 f}{\partial x \partial y}$ & $\frac{\partial^2 f}{\partial y \partial x}$ are equal at a point (x_0, y_0) if

- 1) All first order partial derivatives f_x, f_y ; second order partial derivatives $\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}, \frac{\partial^2 f}{\partial y \partial x}$ exists (here, $n = 2 \implies$ open disc) on an n-ball containing (x_0, y_0) , say B .
- 2) $\frac{\partial^2 f}{\partial x \partial y}$ & $\frac{\partial^2 f}{\partial y \partial x}$ are continuous at (x_0, y_0) . OR
- 2') Either $\frac{\partial^2 f}{\partial x \partial y}$ or $\frac{\partial^2 f}{\partial y \partial x}$ is continuous on B .

2. Chain Rule :-

Theorem 1 (Chain Rule)

Let $w = f(x, y, z)$ be continuous and have continuous first order partial derivatives in a domain D in xyz space. Let

$$x = x(u, v)$$

$$y = y(u, v)$$

$$z = z(u, v)$$

be functions that are continuous & have first order partial derivatives in a domain B in the uv plane, where B is such that for every $(u, v) \in B$, the corresponding point $[x(u, v), y(u, v), z(u, v)]$ lies in D .

[Diagram: Mapping from Region B in the uv -plane containing point (u, v) to Region D in the xyz -space containing point $(x(u, v), y(u, v), z(u, v))$]



Then, the function $w = f(x(u, v), y(u, v), z(u, v))$ is defined in B & has first order partial derivatives w.r.t u & v in B &

$$\left. \begin{aligned} \frac{\partial w}{\partial u} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u}, \\ \frac{\partial w}{\partial v} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v} \end{aligned} \right\} (1)$$

3) Important properties of gradients :-

Let $S \subseteq \mathbb{R}^3$.

Theorem 1: Let $f(x, y, z)$ be a scalar function on S , having continuous first order partial derivatives in S . Then, ∇f exists in S & if $\nabla f(P) \neq \vec{0}$ at some point P , then $\nabla f(P)$ has the direction of maximum rate of increase of f at P .

Theorem 2

Let f be a scalar field $f : S \rightarrow \mathbb{R}$ where S is an open subset of $\mathbb{R}^3$ such that f has continuous first order partial derivatives on S . If the gradient of f at a point P lying on $L(c)$ ($\vec{OP} = \vec{a}$) is not the zero vector, then it is a normal vector of $L(c)$ at P . Equation of the tangent plane at $\vec{a}$

Any point (X, Y, Z) on the tangent plane at $\vec{a}$ satisfies

$$\nabla f(\vec{a}) \cdot (X - a_1, Y - a_2, Z - a_3) = 0$$

or

$$\boxed{\frac{\partial f}{\partial x}(\vec{a})(X - a_1) + \frac{\partial f}{\partial y}(\vec{a})(Y - a_2) + \frac{\partial f}{\partial z}(\vec{a})(Z - a_3) = 0}$$

II. COMPLEX ANALYSIS:-

1) Limit

Theorem 1) Suppose that $f(z) = u(x, y) + iv(x, y)$ where $z = x + iy$, $z_0 = x_0 + iy_0$ & $w_0 = u_0 + iv_0$. Then,

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad \text{iff}$$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} u(x, y) = u_0 \quad \text{and} \quad \lim_{(x, y) \rightarrow (x_0, y_0)} v(x, y) = v_0.$$

Recall: $\lim_{(x, y) \rightarrow (x_0, y_0)} u(x, y) = u_0 \iff \lim_{\|(x, y) - (x_0, y_0)\| \rightarrow 0} \|u(x, y) - u_0\| = 0.$

2) Differentiability and CR equations:-

Theorem 2) Suppose that $f(z) = u(x, y) + iv(x, y)$ & that $f'(z)$ exists at a point $z_0 = x_0 + iy_0$. Then,

- 1) the first order partial derivatives of u & v exist at (x_0, y_0) .
- 2) first order partial derivatives must satisfy the CR eqns

$$u_x = v_y \quad \& \quad u_y = -v_x.$$

- 3) Further,

$$f'(z_0) = u_x + iv_x \Big|_{(x_0, y_0)}$$

Sufficient condn. for differentiability

Theorem 3 : Let $f(z)$ be defined throughout some ε -nbhd of a point $z_0 = x_0 + iy_0$. Suppose that

- a) u_x, u_y, v_x, v_y exist everywhere on the neighbourhood,
- b) u_x, u_y, v_x, v_y are continuous at (x_0, y_0)
- c) u & v satisfy the CR equations at (x_0, y_0)

$$u_x = v_y, \quad u_y = -v_x,$$

then $f'(z_0)$ exists &

$$f'(z_0) = u_x + iv_x \Big|_{(x_0, y_0)}.$$

Theorem 4 (Sufficient condition for differentiability in terms of polar form)

Let the function $f(z) = u(r, \theta) + iv(r, \theta)$ be defined throughout some ε -nbhd of a non-zero point $z_0 = r_0 e^{i\theta_0}$ & suppose that

- a) the first order partial derivatives of the functions u & v with respect to r & θ exist everywhere in the nbhd
- b) those partial derivatives are continuous at (r_0, θ_0) & satisfy the polar form of the CR eqns

$$\begin{aligned} ru_r &= v_\theta \\ u_\theta &= -rv_r \quad \text{at } (r_0, \theta_0). \end{aligned}$$

Then, $f'(z_0)$ exists at $z = z_0$ & its value is

$$f'(z_0) = e^{-i\theta} (u_r + iv_r) \Big|_{(r_0, \theta_0)}.$$

3) Harmonic functions:-

Harmonic Functions

A real valued function $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ is said to be harmonic in a given domain D of the xy plane if

- i) it has continuous partial derivatives of 1st & 2nd order &
- ii) it satisfies the Laplace eqn

$$\nabla^2 h = h_{xx} + h_{yy} = 0 \quad \text{on } D$$

4) Derivatives of all orders:-

If a function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D , then its component functions u & v are harmonic in D .

For this, we assume the following facts :

If $f(z)$ is analytic at z_0 , then all its n^{th} derivatives, $f^{(n)}(z)$ exists at z_0 .

5) Harmonic conjugates:-

Theorem 1 If $f(z) = u(x, y) + iv(x, y)$ is analytic at $z_0 = (x_0, y_0)$, then u & v have continuous partial derivatives of all orders at that point.

Theorem 2 A function $f(z) = u + iv$ is analytic in a domain D iff v is a harmonic conjugate of u .

Defn :- If two given functions u & v are harmonic in a domain D & their first order partial derivatives satisfy CR eqns throughout D then v is said to be a harmonic conjugate of u .

6) M-L Inequality:-

Theorem (M-L inequality)

Let C denote a contour of length L & suppose that a function $f(z)$ is piecewise continuous on C . If M is a non-negative constant such that $|f(z)| \leq M$ for all points z on C at which $f(z)$ is defined, then

$$\left| \int_C f(z) dz \right| \leq ML.$$

III. Mathematics for Machine Learning :-

Matrix Calculus & Jacobians

Jacobian Matrix: If $\mathbf{y} = \mathbf{f}(\mathbf{x})$ where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$, the Jacobian is an $m \times n$ matrix:

$$J = \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

Artificial Neural Networks

Case 3 :- Input 3 neurons & Output 2 neurons

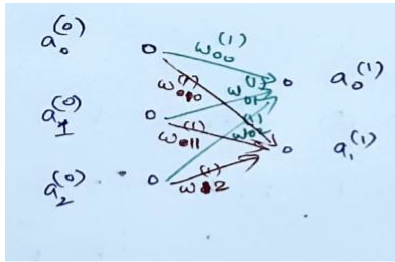
$W_{ij}^{(1)}$ is the weight / line connecting neuron a_i of 1st layer (i.e, $a_i^{(1)}$) & a_j of 0th layer (i.e, $a_j^{(0)}$).

$$a_j^{(0)} \xrightarrow{W_{ij}^{(1)}} a_i^{(1)}$$

$b_0^{(1)}$ is the bias of neuron a_0 in 1st layer.

$b_1^{(1)}$ is the bias of neuron a_1 in 1st layer.

[Diagram: Neural network graph connecting 3 input layer nodes $a_0^{(0)}, a_1^{(0)}, a_2^{(0)}$ to 2 output layer nodes $a_0^{(1)}, a_1^{(1)}$ via weights $W_{ij}^{(1)}$ with biases $b_i^{(1)}$]



$$a_0^{(1)} = \sigma(W_{00}^{(1)} a_0^{(0)} + W_{01}^{(1)} a_1^{(0)} + W_{02}^{(1)} a_2^{(0)} + b_0^{(1)})$$

$$a_1^{(1)} = \sigma(W_{10}^{(1)} a_0^{(0)} + W_{11}^{(1)} a_1^{(0)} + W_{12}^{(1)} a_2^{(0)} + b_1^{(1)})$$

$$\begin{aligned} \vec{a}^{(1)} = \begin{bmatrix} a_0^{(1)} \\ a_1^{(1)} \end{bmatrix} &= \sigma \left(\begin{bmatrix} W_{00}^{(1)} & W_{01}^{(1)} & W_{02}^{(1)} \\ W_{10}^{(1)} & W_{11}^{(1)} & W_{12}^{(1)} \end{bmatrix} \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ a_2^{(0)} \end{bmatrix} + \begin{bmatrix} b_0^{(1)} \\ b_1^{(1)} \end{bmatrix} \right) \\ &= \sigma \left(W^{(1)} \cdot \vec{a}^{(0)} + \vec{b}^{(1)} \right) \end{aligned}$$

Common Activation Functions & Derivatives:

- Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}} \implies \sigma'(z) = ???$
- Tanh: $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \implies \tanh'(z) = ???$

Backpropagation Basics

Recall that the feed-forward equation at the n^{th} layer, $1 \leq n \leq N$ is given by:

$$a^{(n)} = \sigma(z^{(n)})$$

where the **pre-activation variable** $z^{(n)}$ is defined as:

$$z^{(n)} = W^{(n)}a^{(n-1)} + b^{(n)}$$

The variable $z^{(n)}$ represents the state of the neuron after the linear transformation but before the non-linear activation function is applied. The objective is to minimize the **Cost Function** C . For a single training example, we define:

$$C = \frac{1}{2} \|a^{(N)} - y\|^2$$

To update weights W and biases b , we need to determine the gradients $\frac{\partial C}{\partial W}$ and $\frac{\partial C}{\partial b}$.

Example Architecture: Consider a network with an input layer $a^{(0)}$, one hidden layer $a^{(1)}$, and an output layer $a^{(2)}$.

$$a^{(2)} = \sigma(z^{(2)}) \quad \text{and} \quad a^{(1)} = \sigma(z^{(1)})$$

Using the chain rule in **row-vector convention**:

$$\frac{\partial C}{\partial W^{(2)}} = \left(\frac{\partial C}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial z^{(2)}} \right) \cdot \frac{\partial z^{(2)}}{\partial W^{(2)}} \quad (1)$$

$$\frac{\partial C}{\partial W^{(1)}} = \left(\frac{\partial C}{\partial a^{(2)}} \cdot \frac{\partial a^{(2)}}{\partial z^{(2)}} \right) \cdot \frac{\partial z^{(2)}}{\partial a^{(1)}} \cdot \frac{\partial a^{(1)}}{\partial z^{(1)}} \cdot \frac{\partial z^{(1)}}{\partial W^{(1)}} \quad (2)$$

Notice that the term in the parentheses $(\cdot)$ in Eq. (1) is reused in Eq. (2).

Generalized Backpropagation

For a network with N layers, the gradient calculation for the i^{th} layer can be generalized as:

$$\frac{\partial C}{\partial W^{(i)}} = \frac{\partial C}{\partial a^{(N)}} \cdot \frac{\partial a^{(N)}}{\partial z^{(N)}} \cdot \frac{\partial z^{(N)}}{\partial a^{(N-1)}} \cdots \frac{\partial a^{(i+1)}}{\partial z^{(i+1)}} \cdot \frac{\partial z^{(i+1)}}{\partial a^{(i)}} \cdot \frac{\partial a^{(i)}}{\partial z^{(i)}} \cdot \frac{\partial z^{(i)}}{\partial W^{(i)}}$$

IV. Appendix: Some useful Trigonometric Identities :-

Pythagorean identities:

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ \tan^2 \theta + 1 &= \sec^2 \theta \\ 1 + \cot^2 \theta &= \csc^2 \theta\end{aligned}$$

Reciprocal identities:

$$\begin{aligned}\csc x &= \frac{1}{\sin x} \\ \sec x &= \frac{1}{\cos x} \\ \cot x &= \frac{1}{\tan x}\end{aligned}$$

Even - odd identities:

$$\begin{aligned}\sin(-x) &= -\sin x \\ \cos(-x) &= \cos x \\ \tan(-x) &= -\tan x\end{aligned}$$

Product to sum formulas:

$$\begin{aligned}\sin x \cdot \sin y &= \frac{1}{2}[\cos(x-y) - \cos(x+y)] \\ \cos x \cdot \cos y &= \frac{1}{2}[\cos(x-y) + \cos(x+y)] \\ \sin x \cdot \cos y &= \frac{1}{2}[\sin(x+y) + \sin(x-y)] \\ \cos x \cdot \sin y &= \frac{1}{2}[\sin(x+y) - \sin(x-y)]\end{aligned}$$

Sum to product:

$$\begin{aligned}\sin x \pm \sin y &= 2 \sin \left(\frac{x \pm y}{2} \right) \cos \left(\frac{x \mp y}{2} \right) \\ \cos x + \cos y &= 2 \cos \left(\frac{x+y}{2} \right) \cos \left(\frac{x-y}{2} \right) \\ \cos x - \cos y &= -2 \sin \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right)\end{aligned}$$

Double - angle formulas:

$$\begin{aligned}\sin 2\theta &= 2 \sin \theta \cos \theta \\ \cos 2\theta &= \cos^2 \theta - \sin^2 \theta \\ &= 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1 \\ \tan 2\theta &= \frac{2 \tan \theta}{1 - \tan^2 \theta}\end{aligned}$$

Co - function identities:

$$\begin{aligned}\cos \left(\frac{\pi}{2} - x \right) &= \sin x \\ \sin \left(\frac{\pi}{2} - x \right) &= \cos x \\ \tan \left(\frac{\pi}{2} - x \right) &= \cot x \\ \cot \left(\frac{\pi}{2} - x \right) &= \tan x \\ \csc \left(\frac{\pi}{2} - x \right) &= \sec x \\ \sec \left(\frac{\pi}{2} - x \right) &= \csc x\end{aligned}$$

Periodicity identities:

$$\begin{aligned}\sin(x \pm 2\pi) &= \sin x \\ \cos(x \pm 2\pi) &= \cos x \\ \tan(x \pm \pi) &= \tan x \\ \cot(x \pm \pi) &= \cot x \\ \sec(x \pm 2\pi) &= \sec x \\ \csc(x \pm 2\pi) &= \csc x\end{aligned}$$

Sum and difference formulas:

$$\begin{aligned}\sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y \\ \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y \\ \tan(x \pm y) &= \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}\end{aligned}$$

Half - angle formulas:

$$\begin{aligned}\sin \left(\frac{x}{2} \right) &= \pm \sqrt{\frac{1 - \cos x}{2}} \\ \cos \left(\frac{x}{2} \right) &= \pm \sqrt{\frac{1 + \cos x}{2}} \\ \tan \left(\frac{x}{2} \right) &= \frac{1 - \cos x}{\sin x}\end{aligned}$$

Law of sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Law of cosines:

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A \\ A &= \cos^{-1} \left(\frac{b^2 + c^2 - a^2}{2bc} \right)\end{aligned}$$

Area of triangle:

$$\begin{aligned}&\frac{1}{2}ab \sin C \\ &\sqrt{s(s-a)(s-b)(s-c)}, \\ &\text{where } s = \frac{1}{2}(a+b+c)\end{aligned}$$